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1: Library management -system - Identify candidate classes (Books, Readers, Loans, Employees, Publishers) + Books: main entity, book information management.

- + DocGia: person who borrows books.
- + PhieuMuon: records borrowing/returning.
- + NhanVien: system manager (librarian, warehouse keeper...).
- + NhaXuatBan: book source information.

- Properties and behavior (methods) of each class + Book

- o Attributes: maSach (PK, string), nameSach (string), tacGia (string), namXuatBan (int), maNXB (FK), soLuongTon (int), thai page (string: "Available"/"On loan").
- o Behavior: addSach(), capNhatSach(), deleteSach(), saveSach(), capNhatTonKho().

+ DocGia

- o Attributes: maDocGia (PK, string), hoTen (string), daySinh (date), diaChi (string), soDienThoai (string), dayDangKy (date), gioiTinh (boolean).
- o Behavior: dangKyThe(), capNhatThongTin(), viewLichSuMuon(). + PhieuMuon

- o Attributes: maPhieu (PK, string), maDocGia (FK), maSach (FK), maNhanVien (FK), nhaMuon (date), ngoHenTra (date), ngoTraThucTe (date), tinhTrang (string: "On loan"/"Paid"/"Overdue").
- o Behavior: createPhieuMuon(), searchSach(), saveTraQuaHan(), crystalPhieuPhat(). + NhanVien

- o Attributes: maNhanVien (PK, string), hoTen (string), chucVu (string), date of birth (date), soDienThoai (string).
- o Behavior: dangNhap (), quanLyDocGia (), quanLyMuonTra (), quanLySach ().).

+ NhaXuatBan

- o Attributes: maNXB (PK, string), nameNXB (string), diaChi (string), email (string).
- o Behavior: addNXB (), capNhatNXB ().

- Book and Publisher relationships: 1 book

- 1 publisher+

- +Readers and Vouchers: 1 reader – many borrowing slips
- +Books and Borrowing Tickets: 1 book – many borrowing coupons
- +Employees and Loan Vouchers: 1 employee – many vouchers

LESSON 2: Online sales management system

- Identify main use cases

Main actors:

- + Customer (Customers / Potential Customers).
- + NguoiQuanLy / Admin (Administrator / Sales staff).

Main use cases (customer registration, ordering, payment, product management):

KhachHang:

- +DangKyTaiKhan (Customer registration).
- +DangNhap (Sign in).
- +SeeSanPham (Search / View product list).
- +ThemVaoGioHang (Add to cart).
- +DatHang (Order).
- +ThanhToan (Online payment / COD).
- +View DonHang (View order history).

Admin / NguoiQuanLy:

- +QuanLySanPham (Add/Edit/Delete products).
- +QuanLyDonHang (View/Confirm/Cancel order).
- +QuanLyKhachHang (Customer information management).
- +QuanLyThanhToan (Payment processing, refund if necessary).

Extend / Include:

- +ThanhToan includes XacNhanDonHang.
- +DatHang extends ThanhToan (if paying online).

- From use case, find the necessary classes

- +Customers (Customer): from register, view orders.
- +SanPham (Product): product management.
- +GioHang (Cart): temporarily store products before ordering.
- +DonHang (Order): from order, payment.
- +ChiTietDonHang (OrderDetail): product details in the order.
- +ThanhToan (Payment): payment processing (may have subclasses such as OnlinePayment, COD).

+Admin or NguoiQuanLy (inherits from User if there is a user system).

- Analyze the relationship between classes (specify inheritance if any)
 Customers and Customers: 1 customer - many orders DonCall and
 Customer: 1 order - many order details (Composition) Customer Service and
 SanPham: many order details - 1 product DonHang and ThanhToan: 1 order
 - 1 payment User (abstract class) ←inheritance ←KhachHang and Admin
 (KhachHang and Admin are both the same type of User, sharing the same
 account, password, and full name).+++++

Lesson 3: Convert class diagrams into data tables

1. MonHoc table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maMon	Nvarchar(10)	x		x	x
2	nameMon	Nvarchar(50)			x	
3	soTinChiLT	Int			x	
4	soTinChiTH	Int			x	

2. Science Table

No	School name	Data Types	Bind			
			PK	FK	Not NULL	Unique
1	maHoHoc	Nvarchar(10)	x			x
2	tenKhoaHoc	Nvarchar(50)			x	
3	ngayTao	Date			x	